PYTHON

Programming Fundamentals

Day 1 — Getting Started

- Variables
- Data Types
- I/O
- Operators
- Calculator

Instructor-Led Workshop

▯ Today's Agenda

▯ Setup

Python installation & VS Code configuration

▯ Variables

Declaring, naming, and storing data

▯ Data Types

int, float, str, bool — when to use what

▯ Input / Output

print(), input() and user interaction

▯ Operators

Arithmetic, comparison, and logical ops

▯ Mini Project

Build a Calculator Application

▯ Setup — Python & VS Code

▯ Install Python

1. Visit python.org/downloads
2. Download Python 3.x (latest stable)
3. ▯ Check "Add Python to PATH"
4. Click Install Now
5. **Verify: open terminal → type `python --version`**

▯ Configure VS Code

1. Visit code.visualstudio.com
2. Install VS Code
3. Open Extensions (Ctrl+Shift+X)
4. Search & install "Python" by Microsoft
5. Open folder → create hello.py
6. Run with ▶ button or F5

? Variables

A variable is a named container that stores a value in memory. Python variables are dynamically typed — no need to declare a type!

```python
# Creating variables

name = "Alice"

age = 20

gpa = 3.8

is_student = True
```

Naming Rules

- ? Start with letter or underscore
- ? Use snake_case (e.g. my_name)
- ? Case-sensitive: age ≠ Age
- ? Don't start with a number
- ? Avoid reserved keywords (if, for...)

?Memory Model: name → ['Alice'] age → [20] gpa → [3.8]

Data Types

int

Integer

Whole numbers, no decimal

```
age = 20
year = 2024
```

float

Float

Numbers with decimal point

```
price = 9.99
gpa = 3.8
```

str

String

Text in quotes (single or double)

```
name = "Alice"
city = 'Jaipur'
```

bool

Boolean

True or False only

```
is_active = True
found = False
```

Use type() to check: type(age) → <class 'int'> type(name) → <class 'str'>

▯ Input / Output

print() — Output

```
# Simple print
print("Hello, World!")

# With variable
name = "Alice"
print("Hi,", name)

# f-string (modern)
print(f"Age: {age}")
```

```
▯ Output:
Hello, World!
Hi, Alice
Age: 20
```

input() — User Input

```
# input() always returns str
name = input("Enter name: ")

# Convert to number
age = int(input("Enter age: "))
price = float(input("Price: "))

# Use the value
print(f"Hello, {name}!")
```

⚠ input() always returns a string!
Use int() or float() to convert numbers.

Operators

Arithmetic

Op	Meaning	Example	Result
+	Add	5 + 3	8
-	Subtract	9 - 4	5
*	Multiply	3 * 4	12
/	Divide	10 / 3	3.33
//	Floor Div	10 // 3	3
%	Modulus	10 % 3	1
**	Power	2 ** 3	8

Comparison

Op	Meaning	Example	Result
==	Equal	5 == 5	True
!=	Not Equal	5 != 3	True
<	Less Than	3 < 7	True
>	Greater Than	7 > 3	True
<=	Less or Eq	5 <= 5	True
>=	Grtr or Eq	9 >= 7	True

Logical

Op	Meaning	Example	Result
and	Both True	T and F	False
or	Either True	T or F	True
not	Invert	not True	False
	Short Circuit		
and	Stops early	F and ...	False
or	Stops early	T or ...	True

Type Conversion

Code

```
# Implicit (automatic)
x = 5     # int
y = 2.0   # float
z = x + y  # Python auto-converts
print(z)  # 7.0 ← float

# Explicit (you control)
num_str = "42"
num_int = int(num_str)  # → 42
pi_str = str(3.14)    # → '3.14'
val   = float('3.14') # → 3.14
flag  = bool(1)       # → True
```

Conversion Functions

- `int(x)` → Integer
- `float(x)` → Float
- `str(x)` → String
- `bool(x)` → Boolean

⚠ int("3.5") → ERROR Use int(float("3.5"))

🔢 Mini Project — Calculator Application

calculator.py

```
# Simple Calculator
print('=== Calculator ===')

# Get user inputs
num1 = float(input('Enter num 1: '))
num2 = float(input('Enter num 2: '))
op  = input('Operator (+,-,*,/): ')

# Perform calculation
if op == '+': result = num1 + num2
elif op == '-': result = num1 - num2
elif op == '*': result = num1 * num2
elif op == '/' and num2 != 0:
   result = num1 / num2
else:
   result = 'Error!'

print(f'Result: {result}')
```

Concepts Used

✅ Variables

✅ float data type

✅ input() & print()

✅ Type conversion

✅ Arithmetic operators

✅ if/elif/else logic

⚠ Division by zero check

✅ f-strings for output

? Final Question — Exit Ticket

What will be the output of this code?

```
x = 10
y = "5"
print(x + int(y))
```

A. 105

B. 15

C. TypeError

D. "105"

✓ Correct

? Explanation: int("5") converts the string "5" to integer 5. Then 10 + 5 = 15.
If we had written x + y without int(), Python would raise a TypeError (can't add int + str).

? Key Takeaway: Always convert types explicitly before performing operations between different data types.

▯ Assignment Tasks — Day 1

1 Task 1 — Personal Profile Card (Beginner)

Create a Python program that:

1. Asks the user for: name, age, city, and favourite subject.
2. Stores each in an appropriately named variable.
3. Calculates birth year = 2024 − age.
4. Displays a formatted profile card using print() and f-strings.

2 Task 2 — Smart Bill Splitter (Intermediate)

Build a bill-splitting calculator that:

1. Inputs total bill amount, number of people, and tip percentage (as integers/floats).
2. Calculates tip amount, total with tip, and amount per person.
3. Uses all arithmetic operators (+, -, *, /, %) at least once.
4. Rounds result to 2 decimal places using round() and prints a neat summary.

▯ *Deadline: Before next class* ▯ *Submit: .py file named task1_yourname.py and task2_yourname.py*

Day 1 — Complete! ?

What we covered today:

- ? Python installation & VS Code setup
- ? Variables — naming, storage, dynamic typing
- ? Data types — int, float, str, bool
- ? Input / Output — input() and print() with f-strings
- ? Operators — arithmetic, comparison, logical
- ? Type conversion — int(), float(), str(), bool()
- ? **Mini Project — Calculator Application**

See you next class! ?